

TASKS 1–2

Web Scraping & EDA

Collecting, Structuring & Exploring Data for Analysis

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Tools referenced: Python (BeautifulSoup, Scrapy, pandas, seaborn, SciPy) | Octoparse | ParseHub

TASK 2

Exploratory Data Analysis (EDA): Methodology & Workflow

This report documents the exploratory data analysis performed on the datasets collected in Task 1 (the Northfield Outfitters pricing dataset and the HireWell Boards job-listings dataset). It covers framing meaningful questions, examining the structure of the data, identifying trends and anomalies, validating assumptions statistically, and flagging data quality issues for further cleaning.

1. Asking Meaningful Questions

Before touching the data, a short list of guiding questions was drafted so the analysis stayed focused on decisions that mattered rather than open-ended poking around:

- **Northfield Outfitters (pricing):** Which product categories are priced highest, and how much does price vary within a category?
- **HireWell Boards (jobs):** Which locations post the most listings, and do certain job titles cluster in certain regions?
- **Cross-cutting:** Is the data complete and recent enough to support these comparisons, or are there gaps that need to be addressed first?

Framing questions up front made it possible to decide which variables actually mattered and which exploratory steps could be skipped.

2. Exploring the Data Structure

Each dataset was loaded and profiled to understand its shape, variables, and data types before any analysis began:

```
import pandas as pd

df = pd.read_csv("northfield_outfitters_products.csv")
print(df.shape)
print(df.dtypes)
print(df.info())
df.head()
```

This produced a quick inventory of the dataset:

Variable	Data Type	Notes
product_name	object (text)	Free text, mostly unique

category	object (categorical)	8 distinct values
price_usd	float64	Numeric, right-skewed
date_collected	datetime64	Single scrape date range

The same profiling step was repeated for the HireWell Boards dataset (job_title, company, location, date_collected), confirming both datasets were rectangular, consistently typed, and ready for deeper exploration.

3. Identifying Trends, Patterns & Anomalies

Summary statistics and visualizations were used to surface patterns and flag values that looked unusual:

```
import seaborn as sns
import matplotlib.pyplot as plt

df.groupby("category")["price_usd"].describe()

sns.boxplot(data=df, x="category", y="price_usd")
plt.xticks(rotation=45)
plt.title("Price Distribution by Category")
plt.show()
```

- **Trend:** the “Outdoor Footwear” category showed a steadily higher median price than the rest of the catalog.
- **Pattern:** job listings clustered heavily in three metro areas, with the remaining locations contributing only a handful of postings each.
- **Anomaly:** a small number of listings showed a price of \$0.00 or a job posting with no location — both flagged for the data-quality review in Section 5 rather than included in the trend analysis.

4. Testing Hypotheses & Validating Assumptions

Visual patterns were checked against simple statistical tests before being treated as conclusions:

- **Hypothesis:** “Outdoor Footwear is priced higher than the catalog average.” A one-sample t-test on the category’s prices against the overall mean confirmed the difference was statistically significant ($p < 0.05$).
- **Hypothesis:** “Job postings are evenly distributed across locations.” A chi-square goodness-of-fit test rejected this assumption, supporting the clustering observed visually.

```
from scipy import stats

footwear = df[df["category"] == "Outdoor Footwear"]["price_usd"]
t_stat, p_value = stats.ttest_1samp(footwear, df["price_usd"].mean())
print(t_stat, p_value)
```

Validating each visual pattern with a statistical test before drawing conclusions reduced the risk of treating ordinary sampling noise as a meaningful trend.

5. Detecting Data Issues & Problems

The profiling and exploration steps above also surfaced a number of data-quality issues to resolve before the data is used for reporting or modeling:

- **Missing values:** a small share of HireWell Boards rows had a blank location field, identified via `df.isna().sum()`.
- **Invalid values:** a handful of \$0.00 prices in the Northfield Outfitters data, likely placeholder values from out-of-stock listings rather than genuine prices.
- **Duplicates:** a small number of job postings appeared twice under slightly different titles, caught by fuzzy-matching company + location + posting date.
- **Outliers:** a few extreme high prices sat well outside the interquartile range and were reviewed individually rather than dropped automatically.

Each issue was logged with its row count and proposed treatment (impute, drop, or flag for manual review) so the cleaning step in any follow-on analysis has a clear, documented starting point.

6. Summary

Working through the data in this order — meaningful questions, structural profiling, pattern detection, statistical validation, and data-quality review — turned a raw scrape into a dataset that can be trusted for further analysis. The same workflow applies regardless of source: define what you're trying to learn, understand what you actually have, look for patterns, test them rather than assume them, and document what still needs to be cleaned.